

ELECTRIC VEHICLES, BATTERIES AND ALTERNATIVE FUELS

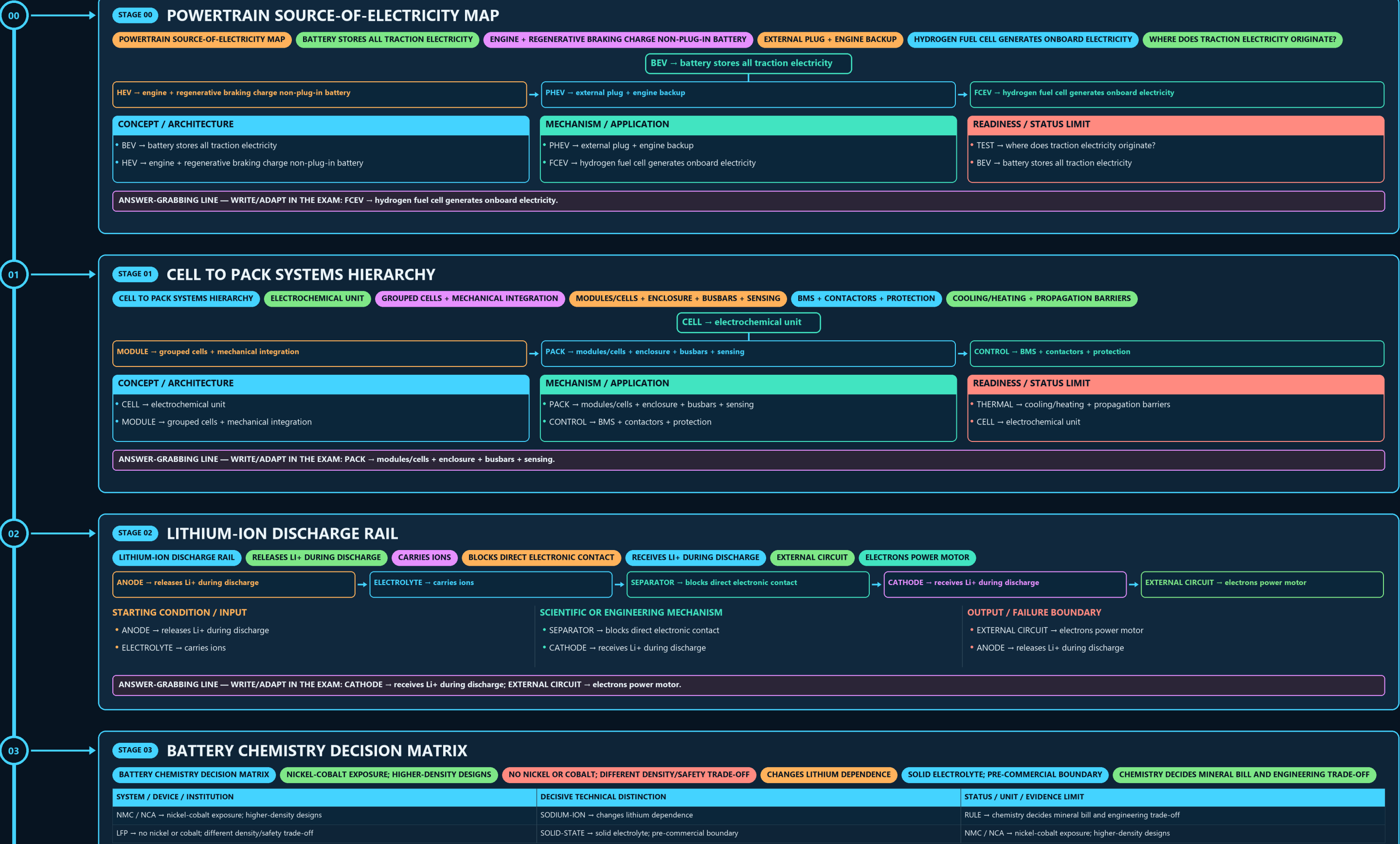
POWERTRAIN SOURCE-OF-ELECTRICITY MAP → CELL TO PACK SYSTEMS → LITHIUM-ION DISCHARGE RAIL → BATTERY CHEMISTRY DECISION MATRIX → ENERGY AND POWER AXES → AC AND DC CHARGING

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g3 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



- ANODE → releases Li+ during discharge
- ELECTROLYTE → carries ions

- SEPARATOR → blocks direct electronic contact
- CATHODE → receives Li+ during discharge

- EXTERNAL CIRCUIT → electrons power motor
- ANODE → releases Li+ during discharge

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CATHODE → receives Li+ during discharge; EXTERNAL CIRCUIT → electrons power motor.

03

STAGE 03

BATTERY CHEMISTRY DECISION MATRIX

BATTERY CHEMISTRY DECISION MATRIX

NICKEL-COBALT EXPOSURE; HIGHER-DENSITY DESIGNS

NO NICKEL OR COBALT; DIFFERENT DENSITY/SAFETY TRADE-OFF

CHANGES LITHIUM DEPENDENCE

SOLID ELECTROLYTE; PRE-COMMERCIAL BOUNDARY

CHEMISTRY DECIDES MINERAL BILL AND ENGINEERING TRADE-OFF

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
NMC / NCA → nickel-cobalt exposure; higher-density designs	SODIUM-ION → changes lithium dependence	RULE → chemistry decides mineral bill and engineering trade-off
LFP → no nickel or cobalt; different density/safety trade-off	SOLID-STATE → solid electrolyte; pre-commercial boundary	NMC / NCA → nickel-cobalt exposure; higher-density designs

SYSTEM / DEVICE / INSTITUTION

- NMC / NCA → nickel-cobalt exposure; higher-density designs
- LFP → no nickel or cobalt; different density/safety trade-off

DECISIVE TECHNICAL DISTINCTION

- SODIUM-ION → changes lithium dependence
- SOLID-STATE → solid electrolyte; pre-commercial boundary

STATUS / UNIT / EVIDENCE LIMIT

- RULE → chemistry decides mineral bill and engineering trade-off
- NMC / NCA → nickel-cobalt exposure; higher-density designs

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → chemistry decides mineral bill and engineering trade-off.

04

STAGE 04

ENERGY AND POWER AXES

ENERGY AND POWER AXES

ENERGY DENSITY

HOW MUCH ENERGY PER MASS/VOLUME

POWER DENSITY

HOW QUICKLY ENERGY IS DELIVERED/ACCEPTED

ENERGY AXIS

FAST CHARGE

POWER AXIS

ENERGY DENSITY → how much energy per mass/volume

POWER DENSITY → how quickly energy is delivered/accepted

RANGE / WEIGHT → energy axis

ACCELERATION / REGEN / FAST CHARGE → power axis

CONCEPT / ARCHITECTURE	MECHANISM / APPLICATION	READINESS / STATUS LIMIT
• ENERGY DENSITY → how much energy per mass/volume • POWER DENSITY → how quickly energy is delivered/accepted	• RANGE / WEIGHT → energy axis • ACCELERATION / REGEN / FAST CHARGE → power axis	• TRAP → one metric cannot certify whole-pack performance • ENERGY DENSITY → how much energy per mass/volume

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → one metric cannot certify whole-pack performance.

05

STAGE 05

AC AND DC CHARGING CONVERSION MAP

AC AND DC CHARGING CONVERSION MAP

GRID AC

VEHICLE ONBOARD CHARGER

REGULATED BATTERY DC

OFFBOARD FAST CHARGER

POWER AND USE CONTEXT

PHYSICAL/ELECTRICAL INTERFACE

COMMUNICATION AND CONTROL

GRID AC → vehicle onboard charger → regulated battery DC

GRID AC → offboard fast charger → regulated battery DC

LEVEL → power and use context

CONNECTOR → physical/electrical interface

STARTING CONDITION / INPUT	SCIENTIFIC OR ENGINEERING MECHANISM	OUTPUT / FAILURE BOUNDARY
• GRID AC → vehicle onboard charger → regulated battery DC • GRID AC → offboard fast charger → regulated battery DC	• LEVEL → power and use context • CONNECTOR → physical/electrical interface	• PROTOCOL → communication and control • GRID AC → vehicle onboard charger → regulated battery DC

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: GRID AC → offboard fast charger → regulated battery DC.

06

STAGE 06

CHARGING GOVERNANCE INSTITUTION MAP

CHARGING GOVERNANCE INSTITUTION MAP

MINISTRY OF POWER

CHARGING GUIDELINES

CONNECTOR AND SAFETY STANDARDS

ELECTRICITY-SYSTEM AND ELECTRICAL SAFETY REGULATION

VEHICLE APPROVAL AND SAFETY

EFFICIENCY ROLE

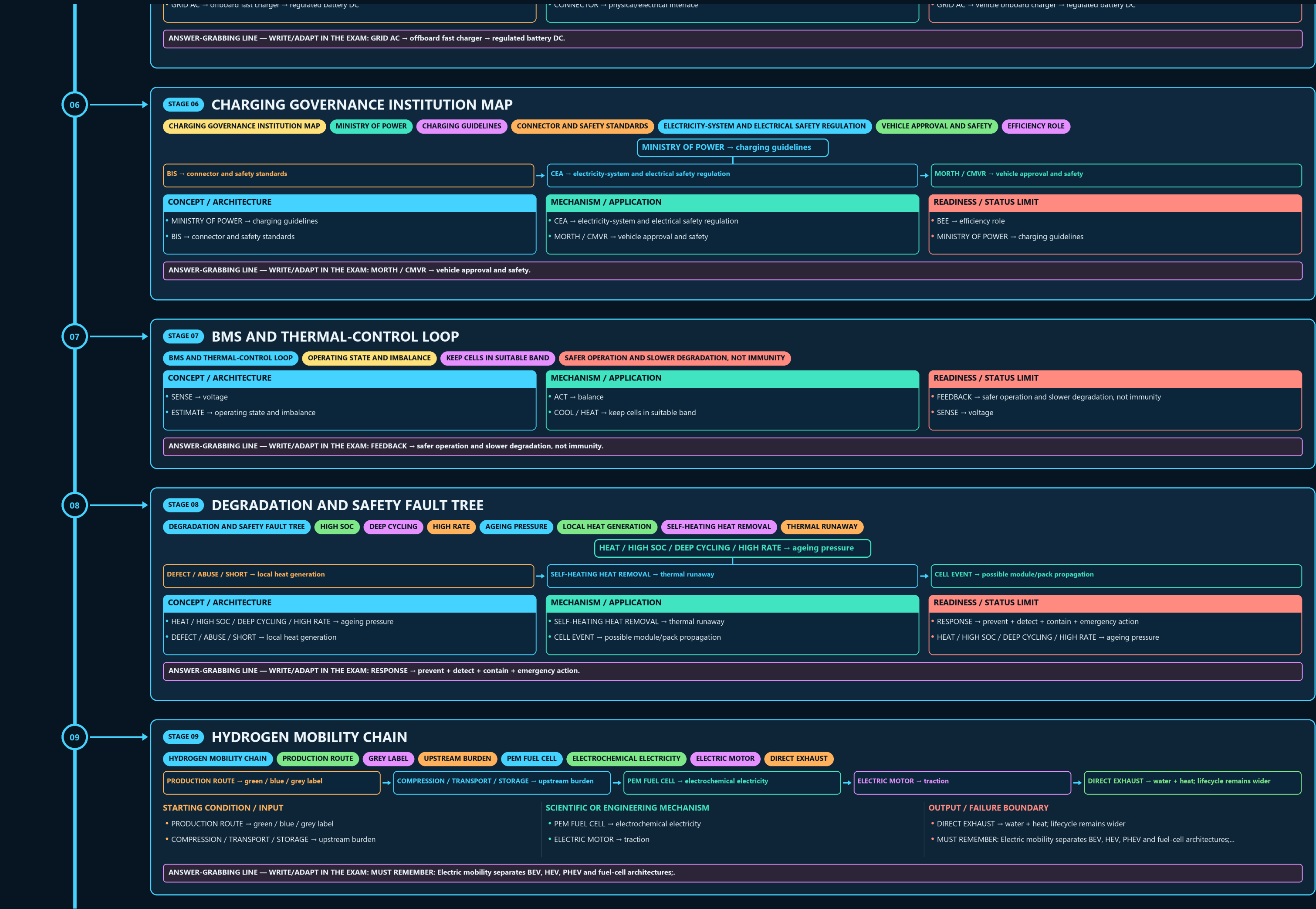
MINISTRY OF POWER → charging guidelines

BIS → connector and safety standards

CEA → electricity-system and electrical safety regulation

MORTH / CMVR → vehicle approval and safety

CONCEPT / ARCHITECTURE	MECHANISM / APPLICATION	READINESS / STATUS LIMIT
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09

STAGE 09 HYDROGEN MOBILITY CHAIN

HYDROGEN MOBILITY CHAIN PRODUCTION ROUTE GREY LABEL UPSTREAM BURDEN PEM FUEL CELL ELECTROCHEMICAL ELECTRICITY ELECTRIC MOTOR DIRECT EXHAUST

PRODUCTION ROUTE → green / blue / grey label → COMPRESSION / TRANSPORT / STORAGE → upstream burden → PEM FUEL CELL → electrochemical electricity → ELECTRIC MOTOR → traction → DIRECT EXHAUST → water + heat; lifecycle remains wider

STARTING CONDITION / INPUT

- PRODUCTION ROUTE → green / blue / grey label
- COMPRESSION / TRANSPORT / STORAGE → upstream burden

SCIENTIFIC OR ENGINEERING MECHANISM

- PEM FUEL CELL → electrochemical electricity
- ELECTRIC MOTOR → traction

OUTPUT / FAILURE BOUNDARY

- DIRECT EXHAUST → water + heat; lifecycle remains wider
- MUST REMEMBER: Electric mobility separates BEV, HEV, PHEV and fuel-cell architectures;...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Electric mobility separates BEV, HEV, PHEV and fuel-cell architectures;

10

STAGE 10 ALTERNATIVE-FUEL PORTFOLIO

ALTERNATIVE-FUEL PORTFOLIO PETROL-BLEND ENGINE PATHWAY DIESEL-BLEND PATHWAY PURIFIED BIOGAS PATHWAY GASEOUS COMBUSTION FUELS HYDROGEN FUEL CELL ELECTROCHEMICAL ELECTRIC-DRIVE PATHWAY CLOSE DISTINCTION

ETHANOL / FLEX-FUEL → petrol-blend engine pathway

BIODIESEL → diesel-blend pathway → CBG / SATAT → purified biogas pathway → CNG / LNG / HCNG → gaseous combustion fuels

CONCEPT / ARCHITECTURE

- ETHANOL / FLEX-FUEL → petrol-blend engine pathway
- BIODIESEL → diesel-blend pathway

MECHANISM / APPLICATION

- CBG / SATAT → purified biogas pathway
- CNG / LNG / HCNG → gaseous combustion fuels

READINESS / STATUS LIMIT

- HYDROGEN FUEL CELL → electrochemical electric-drive pathway
- CLOSE DISTINCTION: Cell is not pack, kW is not kWh, battery capacity is not vehicle...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Cell is not pack, kW is not kWh, battery capacity is not vehicle.

11

STAGE 11 INDIA TRANSITION AND EVIDENCE LADDER

INDIA TRANSITION AND EVIDENCE LADDER END OF LIFE PM E-DRIVE DEMAND/ECOSYSTEM INSTRUMENT PM-EBUS SEWA NGHM PILOTS DISTINCT ROUTES EVIDENCE LIMIT

TAILPIPE → GRID → MANUFACTURE → END OF LIFE → MINERALS → CHEMISTRY → RECYCLING / EPR → PM E-DRIVE → demand/ecosystem instrument → PLI-AUTO / PLI-ACC / PM-eBUS SEWA / NGHM PILOTS → distinct routes → APPROVAL → NOTIFICATION → AWARD → COMMISSIONING → DEPLOYMENT → OUTCOME

STARTING CONDITION / INPUT

- TAILPIPE → GRID → MANUFACTURE → END OF LIFE
- MINERALS → CHEMISTRY → RECYCLING / EPR

SCIENTIFIC OR ENGINEERING MECHANISM

- PM E-DRIVE → demand/ecosystem instrument
- PLI-AUTO / PLI-ACC / PM-eBUS SEWA / NGHM PILOTS → distinct routes

OUTPUT / FAILURE BOUNDARY

- APPROVAL → NOTIFICATION → AWARD → COMMISSIONING → DEPLOYMENT → OUTCOME
- EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State vehicle/fuel pathway, chemistry,...

MECHANISM / VERDICT — EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State vehicle/fuel pathway, chemistry,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State vehicle/fuel pathway, chemistry..

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT THE QUESTION IS NOT "EV VERSUS NO EV"; IT BEVS USUALLY MAKE STRONGEST SENSE WHERE DAILY DRIVING RANGE HYDROGEN MOBILITY IS ANALYTICALLY STRONGER FOR HARD-TO-ELECTRIFY SEGMENTS THAN ROUTINE SMALL PERSONAL TRANSPORT, THOUGH COSTS AND INFRASTRUCTURE REMAIN ETHANOL/CNG/LNG MATTER BECAUSE THE MOBILITY TRANSITION IS BROADER THAN MINISTRY OF HEAVY INDUSTRIES PM E-DRIVE AND ACC BATTERY INDUSTRIAL-POLICY ANCHOR.

OPTIONAL SCIENTIFIC / ENGINEERING DEPTH

- CAUTION: The question is not "EV versus no EV"; it is which powertrain fits which transport segment.
- CAUTION: BEVs usually make strongest sense where daily driving range is moderate and charging access can be organized.
- CAUTION: Hydrogen mobility is analytically stronger for hard-to-electrify segments than for
- routine small personal transport, though costs and infrastructure remain difficult.

READINESS, UNIT OR STATUS LIMIT

- CAUTION: Ethanol/CNG/LNG matter because the mobility transition is broader than battery electrification alone.
- Ministry of Heavy Industries: PM E-DRIVE and ACC battery industrial-policy anchor.
- PM E-DRIVE portal: official implementation interface and public scheme reference.
- ACC battery PLI framework: manufacturing-capacity building, not merely demand subsidy.

COMPARATIVE TECHNOLOGY / GOVERNANCE NUANCE

- Ministry of Power / BEE / BIS / CEA: charging guidelines, standards and power-system interface.
- MNRE: green-hydrogen mission and transport pilot guidelines.
- MoPNG: ethanol-blending and alternative-fuel ecosystem.
- EVs can reduce urban tailpipe emissions and fuel-import vulnerability in selected segments.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.